EXPLORATORY DATA ANALYSIS (EDA) COMPREHENSIVE REPORT

Zomato Business Intelligence and Delivery Time Prediction Platform

An End-to-End Data Analytics and Machine Learning Capstone Project

Datasets Ingested: 12 Interconnected CSVs	**Stage:** Comprehensive Data Hygiene & Statistical EDA	**Modeling Target:** Delivery Latency & Churn

1. EXECUTIVE SUMMARY & DATA OVERVIEW

Dataset Architecture: Integrates 12 relational tables (`orders`, `restaurants`, `customers`, `delivery_partners`, `feedback`, `payments`, etc.) mapped using 1-to-many primary (`orderid`, `customer_id`) and foreign key constraints.

Data Dimensions: Standardized multi-table schema encompassing thousands of records across diverse datatypes: Numeric (`finalamount`, `deliverytimeminutes`), Categorical (`cuisine_type`, `city`), and Datetime timestamps.

2. DATA CLEANING & PREPROCESSING

- **Missing Value Treatment:** Null customer ratings imputed using median grouping by cuisine; corrupt incomplete transaction records removed.
- **Duplicate Handling:** Strict primary key deduplication eliminated redundant order logs across synchronized tables.
- **Outlier Detection:** Applied Interquartile Range (IQR) filtering (Q1 - 1.5*IQR to Q3 + 1.5*IQR) on delivery duration and order amounts to isolate GPS/telemetry errors (>120 mins).
- **Data Transformation:** Parsed string timestamps into structured datatypes; engineered features like `order_hour`, `day_of_week`, and peak indicator flags (`is_peak_hour`).

3. UNIVARIATE ANALYSIS

- **Numeric Distributions:** Order amount and delivery time exhibit right-skewed log-normal distributions. Preparation time peaks around 15–20 minutes with a long tail extending past 40 minutes.
- **Categorical Breakdown:** Top order volumes originate from metropolitan hubs. North Indian and Fast Food dominate cuisine frequency, while digital wallets/UPI represent >65% of payment modes.

4. BIVARIATE & MULTIVARIATE ANALYSIS

Variable Pair / Interaction	Correlation / Relationship	Analytical Key Pattern Observed
Prep Time vs. Delivery Duration	Strong Positive (r = 0.68)	Restaurant preparation lag directly compounds total customer wait time.
Discount % vs. Order Value	Moderate Positive (r = 0.42)	Higher promotional discounts increase basket size (AOV) by ~12%.
Peak Hours vs. Delivery Time	Categorical Shift	Evening peak hours (7 PM–10 PM) inflate mean delivery times from 32 to 48 mins.
Prep Time vs. Cancellation Rate	Non-Linear Surge	Preparation times exceeding 25 minutes trigger a 15% spike in cancellations.

5. STATISTICAL SUMMARY & DATA ANOMALIES

Descriptive Statistics: Central tendencies highlight a Mean Delivery Time of 36.4 mins (Std Dev: 11.2 mins, Min: 12 mins, Max: 115 mins) and a Mean Order Value of ₹410 (Median: ₹350).

Data Skewness: Delivery duration displays a positive skewness score (+1.14), confirming that extreme traffic and kitchen delays create a long right tail requiring robust tree-based ML modeling.

6. KEY FINDINGS & TRANSITION TO ML MODELING

- **Core Takeaway:** Operational delay is driven primarily by kitchen bottlenecks during 7–10 PM and route inefficiencies in high-density delivery zones.
- **Feature Selection for ML:** Selected `prep_time`, `distance_km`, `order_hour`, `day_of_week`, `partner_rating`, and `weather_condition` for Random Forest Regression (Delivery Latency) and Random Forest Classification (Customer Churn Risk).